

Annotating Proteins

Predicting function of proteins.

- [Pfam site docs](#)
- [Pfam manual](#)

Finding homologs

For Protein to Protein searches BLASTP, phmmer ([HMMER](#)), FASTA

```
module load fasta
fasta36 query database > results.FASTA
fasta36 -m 8c -E 1e-3 query database > results.FASTA.tab
```

To Find Domains

See Overview lecture [Domains lecture](#)

Searching with [HMMer](#) against [Pfam](#)

See the HMMER [tutorial](#)

Searching with [Interpro](#)

Searching Interpro on HPCC

Note this can be slow.

```
#SBATCH -p batch -N 1 -n 8
module load iprscan
CPU=4
interproscan.sh --goterms --pathways -f tsv -i PROTEINFILE.fa --cpu $CPU > SEARCH.log
```

The results will contain information like

Gene Ontology <http://geneontology.org/>

Running Analyses on Biocluster

```
module load hmmer
module load db-pfam
hmmsearch --domtblout domtbl_results.out $PFAM_DB/Pfam-A.hmm proteins.fa > proteins.hmmsearch
hmmsearch --domtblout hmmsearch_domtbl_results.out $HMM protein-db.fa > protein.hmmsearch
Pfam2GO - http://current.geneontology.org/ontology/external2go/pfam2go
```

Workshop in class

Let's compare the genomes of cyanobacteria and identify if there are differences in gene content based on Protein domains.

Many papers investigating the evolution and genomes of cyanobacteria.

- <https://journals.asm.org/doi/10.1128/mbio.00561-19>
 - <https://bmcecolevol.biomedcentral.com/articles/10.1186/1471-2148-10-24>
1. Searching for Pfam domains in a set of proteins from several species (start with 3)
 - Let's download some genomes/proteomes from [NCBI](#)
 2. Parsing report files and count the number of domains per species (in python)
 3. Summarize the content comparing with a table sorted by counts